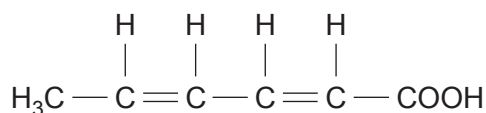


Section BAnswer **all** questions.

8. (a) Sorbic acid was isolated in 1859 from mountain ash berry oil. It is a white solid that is slightly soluble in cold water.



- (i) The solubility of sorbic acid in water is 1.6 g dm^{-3} at 20°C and 40.0 g dm^{-3} at 100°C .

Calculate how much sorbic acid is precipitated from its aqueous solution if 200 cm^3 of a saturated solution at 100°C is cooled to 20°C .

Give your answer to an appropriate number of significant figures.

[2]

.....g

- (ii) Sorbic acid and its salts, for example sodium sorbate, have important uses as antimicrobial agents in food preservation.
Some moulds are, however, able to detoxify the action of these sorbates.
An example is the decarboxylation of sodium sorbate.

I. State what is meant by 'decarboxylation'.

[1]

.....

- II. In the laboratory sodium sorbate can be decarboxylated by heating it with soda lime.

The organic product of decarboxylation is *E*-penta-1,3-diene.

Write the equation for this decarboxylation of sodium sorbate with soda lime (which you should represent as NaOH in your equation), showing the structure of *E*-penta-1,3-diene.

[2]

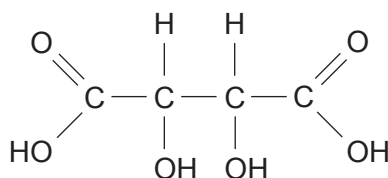


- III. *E*-penta-1,3-diene can then react with hydrogen using platinum as a catalyst giving pentane.

State the number of moles of hydrogen required to react with 0.2 mol of *E*-penta-1,3-diene in this way. [1]

.....

- (b) Tartaric acid (2,3-dihydroxybutanedioic acid) is used in the food industry.



- (i) Indicate any chiral centre(s) present by means of an asterisk (*). [1]

- (ii) This acid occurs in a number of optically active forms.
Complete the sentences below.

Forms of tartaric acid that rotate the plane of plane-polarised light are called

..... [1]

A solution containing an equimolar mixture of two forms of the acid that rotates the plane of plane-polarised light in equal and opposite directions is called a

..... [1]

- (c) Both sorbic acid and tartaric acid are described as showing stereoisomerism.

Explain what is meant by the term stereoisomerism. [1]

.....

.....

- (d) Tartaric acid is formed when butenedioic acid reacts with a suitable oxidising agent.

Give the equation for this reaction, representing the oxidising agent as [O] and using water as a reactant. [1]



- (e) The solubility of unsaturated carboxylic acids in water decreases as the chain length increases.

Number of carbon atoms in the alkyl chain	Solubility in water at 25 °C / g dm ⁻³
3	94
5	1.6
7	0.7

Suggest why this solubility decreases as the chain length increases.

[2]

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